OCR J277 Paper 1: Computer systems

Dougie, these notes cover everything in the OCR specification for Paper 1 of the GCSE (J277/01, Computer systems). The exam lasts 1 hour 30 minutes, is worth 80 marks and counts for 50% of your GCSE. You are not allowed a calculator. All questions are compulsory, and the paper has multiple choice questions, short and medium answer questions, and one 8-mark extended response question. Each section below starts with its code (such as 1.1a) and covers a part of the specification. Read "What you need to know" first, and try the worked examples yourself before reading the answer. Learn the "Key terms" until you can give each meaning without looking. Read "Exam mistakes that lose marks" before every practice paper. Finish each section with the "Quick check": cover the answers, write your own, then mark them.

How to answer exam questions

OCR uses the same command words in every paper. Each one tells you what kind of answer gets the marks.

Command word	What OCR means	What you should do
State	Give a specific name, value or other brief answer without explanation or calculation.	Write the fact in a few words. No need to explain.
Describe	Give a detailed account or picture of a situation, event, pattern or process.	Say what something is or what happens, step by step. You do not need to say why.
Explain	Give a detailed account including reasons or causes.	Make a point, then say why or how. Linking words help: "because", "so", "which means".
Compare	Give an account of the similarities and differences between two (or more) items or situations, referring to both (all) of them throughout.	Write about both things in every sentence: "An SSD is faster than an HDD because...". Two separate lists do not count as comparing.
Evaluate	Assess the implications and limitations. Make judgements about the ideas, works, solutions or methods in relation to selected criteria.	Give points for and against, then make a judgement and back it up.
Discuss	Offer a considered and balanced review that includes a range of arguments, factors or hypotheses.	Cover several sides of the issue, support each point, and give a clear conclusion.

Extended response question. Paper 1 has one 8-mark extended response question. It checks whether you can build a sustained line of reasoning, so it is marked on the quality of the whole answer, not by counting facts. To do well:

- Spend a minute planning. Jot down the points you will make under the headings the question gives you.
- Cover a range of points, not one point repeated.
- Explain each point fully and link it to the scenario in the question.
- Give both sides where the question asks you to discuss or evaluate.
- Finish with a conclusion that answers the question directly.

Calculation questions: you will not have a calculator, so the numbers are chosen to work out neatly. Always show your working and write the units in your answer.

1.1a The CPU and the fetch-execute cycle

Spec point 1.1.1 Architecture of the CPU

What you need to know

The purpose of the CPU

The CPU (central processing unit) processes the data and instructions that make a computer work. It does this by running the fetch-execute cycle over and over: it fetches an instruction from memory, decodes it to work out what it means, then executes (carries out) it. Modern CPUs do this billions of times every second.

Von Neumann architecture

Most computers use Von Neumann architecture. Its main idea is that program instructions and data are stored together in the same memory, and the CPU fetches them from that memory one at a time. Von Neumann architecture uses a set of special registers to do this: the MAR, the MDR, the Program Counter and the Accumulator.

Common CPU components

Component	What it does	Its job during the fetch-execute cycle
ALU (Arithmetic Logic Unit)	Carries out arithmetic (such as adding and subtracting) and logic operations (such as comparing two values, or AND, OR and NOT). It also carries out binary shifts.	Does the calculations and comparisons during the execute stage. Results go into the Accumulator.
CU (Control Unit)	Decodes instructions and sends out control signals to manage the rest of the CPU. It controls how data moves around the CPU and between the CPU and memory or other devices.	Controls the whole cycle. It decodes each instruction and tells the other parts what to do.
Cache	A small amount of very fast memory inside or very close to the CPU. It stores data and instructions that are used often, so the CPU does not have to wait for slower RAM.	The CPU checks the cache first when it needs an instruction or data. If it is there, it is fetched much faster than from RAM.
Registers	Tiny, extremely fast storage locations inside the CPU. Each one holds a single item: an instruction, a piece of data or a memory address.	Hold the addresses, instructions and data the CPU is working on right now.

The registers you need to know

Register	What it stores	Data or address?
Program Counter (PC)	The address of the next instruction to be fetched. It goes up by 1 each cycle so the CPU moves through the program.	Address
MAR (Memory Address Register)	The address of the memory location that is about to be read from or written to.	Address
MDR (Memory Data Register)	The data or instruction that has just been fetched from memory, or that is about to be written to memory.	Data (an instruction counts as data here)
Accumulator	The result of calculations done by the ALU, while it is being used.	Data

Storing data versus storing an address

Memory is made of numbered locations. The number of a location is its **address**. The value kept in that location is the **data**. Think of a street: the house number is the address, and whoever lives in the house is the data. If memory location 25 holds the value 7, then 25 is the address and 7 is the data. The PC and MAR hold addresses. The MDR and Accumulator hold data.

The fetch-execute cycle

Fetch

1. The address held in the Program Counter is copied into the MAR.
2. The CPU fetches the instruction stored at that address in memory, and it is copied into the MDR.
3. The Program Counter goes up by 1, so it now holds the address of the next instruction.

Decode

4. The Control Unit decodes the instruction to work out what it has to do.

Execute

5. The instruction is carried out. Depending on the instruction, this might mean fetching data from memory (its address goes into the MAR and the data arrives in the MDR), the ALU doing a calculation and putting the result in the Accumulator, or writing data back to memory.

The cycle then starts again with the next instruction.

You need to know what happens at each stage. OCR does not require you to know every detail of how data passes between the registers at each stage.

Worked example: one trip round the cycle

The Program Counter holds 20. Memory location 20 holds the instruction "add the number stored in location 51 to the Accumulator". Location 51 holds the number 6, and the Accumulator holds 4.

1. Fetch: 20 is copied from the PC into the MAR. The instruction at address 20 is fetched into the MDR. The PC goes up to 21.
2. Decode: the CU decodes the instruction and finds it is an "add" that needs the data at address 51.
3. Execute: 51 goes into the MAR, and the number 6 is fetched into the MDR. The ALU adds 6 to the 4 in the Accumulator. The Accumulator now holds 10.
4. The cycle starts again, fetching the instruction at address 21.

Key terms

CPU (central processing unit)
The part of the computer that processes data and instructions by running the fetch-execute cycle.

Von Neumann architecture
A computer design where instructions and data are stored together in the same memory and fetched by the CPU one at a time.

Fetch-execute cycle
The repeated process of fetching an instruction from memory, decoding it and executing it.

Fetch
The stage where the next instruction is copied from memory into the CPU.

Decode
The stage where the Control Unit works out what the instruction means.

Execute
The stage where the instruction is carried out.

ALU (Arithmetic Logic Unit)
The part of the CPU that does arithmetic, logic operations, comparisons and binary shifts.

CU (Control Unit)
The part of the CPU that decodes instructions and sends control signals to manage the fetch-execute cycle.

Cache
Small, very fast memory in or near the CPU that holds frequently used data and instructions.

Register
A tiny, extremely fast storage location inside the CPU that holds one instruction, piece of data or address.

Program Counter (PC)
The register that holds the address of the next instruction to be fetched.

MAR (Memory Address Register)
The register that holds the address of the memory location about to be read from or written to.

MDR (Memory Data Register)
The register that holds the data or instruction just fetched from memory, or about to be written to it.

Accumulator
The register that holds the results of calculations carried out by the ALU.

Address
The number that identifies a location in memory.

Data
The value stored at a memory location.

Exam mistakes that lose marks

- **Mixing up the MAR and the MDR.** The MAR holds an *address*; the MDR holds *data*. Use the letters: MAR has A for Address, MDR has D for Data.
- **Saying the Program Counter counts how many instructions have run.** It holds the *address of the next instruction*.
- **Saying the Accumulator stores addresses.** It stores the results of ALU calculations, which are data.
- **Saying the CU does the calculations.** The ALU does calculations and comparisons. The CU decodes instructions and controls everything else.
- **Giving only "fetch, decode, execute" when asked to describe the cycle.** Say what happens at each stage: PC to MAR, instruction into MDR, PC goes up by 1, CU decodes, instruction carried out.
- **Describing cache as large or as a type of storage for files.** Cache is small, very fast memory that holds frequently used data and instructions.
- **Treating registers and RAM as the same thing.** Registers are inside the CPU and each holds only one item. RAM is separate from the CPU and holds many programs and their data.

Quick check

1. State what the Program Counter holds.
2. Which register holds the data that has just been fetched from memory?
3. Name the CPU component that decodes instructions.
4. Describe what happens during the fetch stage of the fetch-execute cycle.
5. Explain the difference between an address and data.

▼ **Answers**

1. The address of the next instruction to be fetched.
2. The MDR (Memory Data Register).
3. The CU (Control Unit).
4. The address in the Program Counter is copied into the MAR. The instruction at that address is fetched from memory into the MDR. The Program Counter goes up by 1.

5. An address is the number of a location in memory; data is the value stored at that location. For example, location 25 (the address) might hold the value 7 (the data).

1.1b CPU performance and embedded systems

Spec points 1.1.2 CPU performance and 1.1.3 Embedded systems

What you need to know

Clock speed

Clock speed is the number of fetch-execute cycles the CPU can carry out each second. It is measured in hertz (Hz). 1 Hz is one cycle per second, so a 3.5 GHz (gigahertz) CPU can carry out 3.5 billion cycles each second.

- A higher clock speed means more instructions are executed each second, so programs run faster.
- A higher clock speed also uses more power and produces more heat.

Cache size

Cache is very fast memory in or near the CPU. Cache size is how much data and how many instructions it can hold.

- A larger cache can hold more of the data and instructions the CPU uses often.
- That means the CPU has to fetch from slower RAM less often, so it spends less time waiting and runs faster.
- Cache is expensive, so CPUs only have a small amount.

Number of cores

A core is a processing unit inside the CPU that can fetch, decode and execute instructions on its own. A dual-core CPU has two cores; a quad-core CPU has four.

- More cores means more instructions can be processed at the same time, which helps when running several programs at once (multitasking) or programs designed to split up their work.
- Doubling the cores does *not* always double the speed. The software has to be written to use more than one core, some tasks must happen in order and cannot be split up, and the cores spend some time sharing out the work between them.

Changing the characteristics, alone or together

Change	Effect on performance
Increase clock speed	More instructions per second, so faster. More heat and power used.
Increase cache size	Fewer slow fetches from RAM, so faster, especially for data and instructions that are used again and again.
Increase number of cores	More instructions processed at the same time, so faster, but only for software that can use the extra cores.
Increase all three	Biggest improvement overall, but the real gain still depends on the software being run.

Worked example: comparing two CPUs

CPU A: 4 cores, 3.0 GHz, 8 MB cache. CPU B: 2 cores, 3.6 GHz, 4 MB cache.

- For video editing software that is written to use many cores, CPU A is likely to be faster. It can process twice as many instructions at the same time, and its larger cache means fewer waits for RAM.
- For an older program that can only use one core, CPU B might be faster. Each of its cores carries out 3.6 billion cycles per second compared with 3.0 billion on CPU A. However, CPU A's bigger cache helps it, so the gap may be small.
- Conclusion: performance depends on all the characteristics together and on the software being run, not on one number alone.

Embedded systems

An embedded system is a computer system built into a larger device, where it does one dedicated job. A general-purpose computer, like a laptop, can run lots of different programs. An embedded system usually runs just one program, designed for the device it is part of.

Typical characteristics of embedded systems:

- Do one dedicated task (or a small set of tasks).
- Small in size, so they fit inside the device.
- Use little power, so they can run on batteries or run all the time cheaply.
- Cheap to make, because they are mass-produced and only need simple hardware.
- Very reliable, because they do one tested job.
- The program is usually stored in ROM, so it cannot easily be changed by the user.
- Often have a very simple user interface (a few buttons or a small display), or none at all.

Examples of embedded systems: washing machines, dishwashers, microwave ovens, central heating thermostats, traffic lights, car engine management systems, car satnavs, digital cameras, digital watches, burglar alarms, smart TVs and medical equipment such as heart rate monitors.

Worked example: is it an embedded system?

A washing machine has a computer that controls the water, heating and motor. It only runs the washing programs, is built into the machine and is controlled with a dial and buttons. It is an embedded system. A laptop runs many different programs that the user chooses and installs, so it is a general-purpose computer, not an embedded system.

Key terms

Clock speed
The number of fetch-execute cycles a CPU can carry out per second.

Hertz (Hz)
The unit of clock speed: one cycle per second (1 GHz is one billion cycles per second).

Cache size
How much data and how many instructions the CPU's cache can hold.

Core

A processing unit in a CPU that can fetch, decode and execute instructions on its own.

Multi-core processor
A CPU with more than one core, such as dual-core (two) or quad-core (four).

Performance
How quickly a CPU can process instructions.

Embedded system
A computer system built into a larger device to carry out one dedicated function.

General-purpose computer
A computer, such as a laptop or desktop, that can run many different programs chosen by the user.

Exam mistakes that lose marks

- **Saying more cores always means double the speed.** Say that software must be written to use the extra cores, and some tasks cannot be split up.
- **Defining clock speed as "how fast the computer is".** Say it is the number of fetch-execute cycles (instructions) per second, measured in hertz.
- **Thinking cache size is how much storage the computer has.** Cache is a small amount of fast memory for data and instructions the CPU uses often; it has nothing to do with saving files.
- **Not explaining why a larger cache helps.** Say that the CPU has to fetch from slower RAM less often.
- **Giving a smartphone, tablet or laptop as an embedded system.** These are general-purpose computers. Use a washing machine, microwave or traffic light instead.
- **Listing characteristics of embedded systems with no reason.** In an "explain" question, add why: "low power, so it can run from a small battery".
- **Judging a CPU on one number.** When comparing, mention clock speed, cache and cores together, and link to the task.

Quick check

1. State what clock speed measures.
2. Explain why a larger cache can improve CPU performance.
3. Give two reasons why doubling the number of cores might not double performance.
4. Define an embedded system.
5. Give three characteristics of embedded systems.

▼ **Answers**

1. The number of fetch-execute cycles the CPU can carry out per second, measured in hertz.
2. More frequently used data and instructions can be kept in the cache, so the CPU fetches from slower RAM less often and spends less time waiting.
3. Any two: the software may not be written to use more than one core; some tasks have to be done in order and cannot be split; the cores spend time sharing out the work.
4. A computer system built into a larger device that carries out one dedicated function.

5. Any three: one dedicated task; small; low power; cheap to make; reliable; program stored in ROM that is hard to change; simple or no user interface.

1.2a Primary and secondary storage

Spec points 1.2.1 Primary storage (memory) and 1.2.2 Secondary storage

What you need to know

The need for primary storage

Primary storage (memory) is storage that the CPU can access directly. Computers need it because the CPU can only work on instructions and data held in primary storage. Secondary storage is far too slow for the CPU to work from directly, so programs and data are copied into primary storage while they are being used. Primary storage usually consists of RAM and ROM.

RAM (random access memory)

- **Purpose:** holds the operating system, the programs that are running and the data they are using right now.
- **Volatile:** everything in RAM is lost when the power is turned off.
- Can be read from and written to, so its contents change all the time.
- Larger than ROM (usually several gigabytes in a modern computer).

ROM (read only memory)

- **Purpose:** holds the instructions the computer needs to start up (boot up) when it is switched on.
- **Non-volatile:** keeps its contents when the power is turned off.
- Read only: its contents cannot normally be changed.
- Very small compared with RAM.

	RAM	ROM
Volatile?	Yes: contents lost when power is off	No: contents kept when power is off
Read and write?	Read and write	Read only
What it holds	Operating system, running programs and their data	Start-up (boot-up) instructions
Size	Larger (gigabytes)	Small

Virtual memory

Why it may be needed: if RAM is full and another program or more data needs to be loaded, the computer needs somewhere to put the extra data.

How it works:

1. The operating system sets aside part of secondary storage (such as the hard drive) to act as extra memory. This is virtual memory.
2. When RAM is full, data that is not being used right now is moved from RAM to virtual memory, which frees up space in RAM.
3. When that data is needed again, it is moved back from virtual memory into RAM (and other data may be moved out to make room).

Virtual memory lets more programs run than would fit in RAM, but secondary storage is much slower than RAM, so the computer slows down when it uses virtual memory a lot. Adding more RAM means virtual memory is needed less.

Cache

Cache is also primary storage. It is very fast, very small memory in or next to the CPU that holds frequently used data and instructions. It is faster than RAM but much more expensive, so there is only a little of it. From fastest to slowest: registers, cache, RAM, secondary storage.

The need for secondary storage

Secondary storage is non-volatile storage that keeps files, programs and the operating system when the power is off. Computers need it because RAM is volatile (it loses everything when switched off) and is too small and expensive to hold everything. Secondary storage holds data long term, in much larger amounts, at a lower cost per gigabyte.

Common types of secondary storage

Type	Examples	How it stores data (in short)
Optical	CD, DVD, Blu-ray disc	A laser reads data from the surface of a spinning disc.
Magnetic	Hard disk drive (HDD), magnetic tape	Data is stored magnetically on spinning disks (HDD) or on a long strip of tape.
Solid state	Solid state drive (SSD), USB flash drive (memory stick), SD card	Data is stored electronically in flash memory chips, with no moving parts.

You do not need to know the parts inside each type of storage.

The characteristics used to compare storage

- **Capacity:** how much data it can hold.
- **Speed:** how quickly data can be read from it and written to it.
- **Portability:** how easy it is to carry around (its size and weight).
- **Durability:** how well it survives being dropped, knocked, scratched or heated.
- **Reliability:** how likely it is to keep working and keep data safe over time and over many uses.
- **Cost:** how much it costs, usually compared as cost per gigabyte.

	Magnetic (HDD)	Solid state (SSD, flash)	Optical (CD, DVD, Blu-ray)
Capacity	High (often several terabytes)	Medium to high (less than an HDD for the same price)	Low (CD about 700 MB, DVD about 4.7 GB, Blu-ray about 25 GB)
Speed	Medium: moving parts slow it down	Very fast: no moving parts	Slow
Portability	Portable versions exist but are heavier and bulkier	Very portable: small and light	Very portable: thin and light, but needs a disc drive
Durability	Moving parts can be damaged if dropped, and strong magnets can damage data	Very durable: no moving parts, survives being dropped	Easily scratched or snapped
Reliability	Good, but moving parts wear out over time	Very reliable, though each memory cell can only be written to a limited number of times	Data can be lost if the disc is scratched or ages
Cost	Cheap per GB	More expensive per GB	Very cheap per disc

Magnetic tape has a very high capacity and a very low cost per GB, but data has to be read in order along the tape, so it is slow to find a file. Organisations use it for backups.

Choosing storage for a given use

To choose storage, work out which characteristics are most important in the scenario, then pick the type that does best on those. Always give a reason linked to the scenario.

Worked example 1: a travelling photographer

A photographer travels around the country and needs to carry about 2 TB of photos in a bag.

- Important characteristics: capacity (2 TB), portability, durability (it will be carried and may be knocked).
- Choice: a portable (external) SSD.
- Reasons: it is small and light, so easy to carry; it has no moving parts, so it is unlikely to be damaged if the bag is dropped; it has enough capacity for 2 TB; it is fast, so photos copy quickly.
- Drawback: it costs more per GB than an HDD, but the durability is worth it for someone who travels.

Worked example 2: a school backup

A school needs to back up 30 TB of data every night and keep the backups in a locked room.

- Important characteristics: capacity and cost. Speed of finding a single file is less important, and the backups do not need to be carried around.
- Choice: magnetic storage (magnetic tape or hard disk drives).
- Reasons: very high capacity at a low cost per GB. Portability and durability are less important because the backups stay in one place.

Key terms

Primary storage (memory)
Storage the CPU can access directly, such as RAM, ROM and cache.

Secondary storage
Non-volatile storage used to keep files, programs and the operating system long term.

RAM (random access memory)
Volatile, read and write memory that holds the operating system, running programs and their data.

ROM (read only memory)
Non-volatile, read only memory that holds the start-up instructions.

Volatile
Loses its contents when the power is turned off.

Non-volatile
Keeps its contents when the power is turned off.

Boot up
The process of starting a computer when it is switched on, using instructions stored in ROM.

Virtual memory
Part of secondary storage used as extra memory when RAM is full.

Cache
Very fast, small memory in or near the CPU that holds frequently used data and instructions.

Optical storage
Storage on discs read by a laser, such as CDs, DVDs and Blu-ray discs.

Magnetic storage
Storage that records data magnetically, such as hard disk drives and magnetic tape.

Solid state storage
Storage that uses flash memory chips with no moving parts, such as SSDs, USB flash drives and SD cards.

Storage device
The hardware that reads and writes data, such as a DVD drive or hard disk drive.

Storage media
The thing the data is actually stored on, such as a DVD or a tape.

Capacity
How much data a storage device can hold.

Speed
How quickly data can be read from or written to storage.

Portability
How easy a storage device is to carry around.

Durability
How well a storage device survives physical damage such as drops and scratches.

Reliability
How likely a storage device is to keep working and keep data safe over time.

Cost
The price of storage, usually compared as the cost per gigabyte.

Exam mistakes that lose marks

- **Confusing RAM with storage.** "My phone has 128 GB of memory" is about secondary storage. RAM holds programs and data in use and is wiped when the power goes off; storage keeps files.
- **Saying ROM stores the user's files or the whole operating system.** ROM holds the start-up instructions. The operating system is kept in secondary storage and loaded into RAM.
- **Saying RAM is non-volatile, or ROM is volatile.** RAM is volatile; ROM is non-volatile.
- **Saying virtual memory makes the computer faster.** It lets more programs run when RAM is full, but it is slower than RAM, so performance drops.
- **Calling virtual memory "extra RAM".** It is part of secondary storage that is used as if it were memory.
- **Writing "SSD is better" without a reason.** Pick the characteristic and link it to the scenario: "an SSD has no moving parts, so it will not break if the student drops their bag".
- **Saying "cheap" or "expensive" without saying per GB.** A large HDD can cost more in total than a small SSD, but less per GB.
- **Mixing up durability and reliability.** Durability is surviving physical damage; reliability is working correctly over time.
- **Saying optical discs or SSDs are magnetic.** Optical uses a laser; solid state uses flash memory chips.

Quick check

1. State two differences between RAM and ROM.
2. Explain why computers need secondary storage.
3. Describe how virtual memory works.
4. Give one example each of optical, magnetic and solid state storage.
5. A student needs storage to carry coursework between home and school. Recommend a storage type and give two reasons.

▼ **Answers**

1. Any two: RAM is volatile but ROM is non-volatile; RAM can be read and written but ROM is read only; RAM holds running programs and data but ROM holds the start-up

instructions; RAM is larger than ROM.

2. RAM is volatile, so files, programs and the operating system would be lost when the power is off. Secondary storage is non-volatile and keeps them long term, with a large capacity at low cost.
3. When RAM is full, the operating system moves data that is not currently needed from RAM to an area of secondary storage. When that data is needed again it is moved back into RAM.
4. Optical: CD, DVD or Blu-ray. Magnetic: hard disk drive or magnetic tape. Solid state: SSD, USB flash drive or SD card.
5. Solid state (a USB flash drive). Any two reasons: small and light so very portable; no moving parts so durable if dropped; enough capacity for coursework; low cost for a small capacity.

1.2b Units and number systems

Spec points 1.2.3 Units and 1.2.4 Data storage (numbers)

What you need to know

Why data is stored in binary

A computer is made of billions of tiny switches (transistors) that can only be in one of two states: on or off. These two states are written as 1 and 0, which is binary. So all data (numbers, text, images, sound and program instructions) has to be converted into binary before a computer can store or process it.

Units of data storage

Unit	Size
Bit	A single 1 or 0 (the smallest unit)
Nibble	4 bits
Byte	8 bits
Kilobyte (KB)	1,000 bytes
Megabyte (MB)	1,000 KB
Gigabyte (GB)	1,000 MB
Terabyte (TB)	1,000 GB
Petabyte (PB)	1,000 TB

To go to a **bigger** unit, divide (by 8 from bits to bytes, then by 1,000 each step). To go to a **smaller** unit, multiply (by 1,000 each step, then by 8 from bytes to bits). OCR also accepts 1,024 instead of 1,000. Pick one and use it all the way through a question.

> **Worked example: moving between units**
>
> - 3.5 GB to MB: 3.5 × 1,000 = 3,500 MB.
> - 2 MB to bytes: 2 × 1,000 × 1,000 = 2,000,000 bytes.
> - 64,000 bits to KB: 64,000 ÷ 8 = 8,000 bytes; 8,000 ÷ 1,000 = 8 KB.
> - 4,500 GB to TB: 4,500 ÷ 1,000 = 4.5 TB.
> - 3 bytes to nibbles: 3 × 8 = 24 bits; 24 ÷ 4 = 6 nibbles.

Data capacity and calculating storage needs

Storage devices have fixed capacities, such as a 32 GB USB flash drive or a 2 TB hard drive. To check whether files will fit, convert every size to the same unit, add them up, then compare with the capacity.

Worked example: will the files fit?

Dougie wants to put 300 photos of 5 MB each and 20 videos of 500 MB each on a 16 GB USB flash drive.

- Photos: 300 × 5 = 1,500 MB.
- Videos: 20 × 500 = 10,000 MB.
- Total: 1,500 + 10,000 = 11,500 MB = 11.5 GB.
- 11.5 GB is less than 16 GB, so the files fit, with 4.5 GB left over.

How many 4 MB songs fit on a 2 GB drive? 2 GB = 2,000 MB; 2,000 ÷ 4 = 500 songs.

File size formulas

Each formula gives the answer in **bits**. Divide by 8 to get bytes, then by 1,000 for KB, and so on.

- **Sound file size** = sample rate (Hz) × duration (seconds) × bit depth
- **Image file size** = colour depth × image height (pixels) × image width (pixels)
- **Text file size** = bits per character × number of characters

Real files also hold metadata (extra information about the file). OCR accepts answers that add an allowance for metadata, but the formulas above are what you need.

Worked example: image file size

An image is 200 pixels wide and 100 pixels high, with a colour depth of 4 bits.

- 4 × 100 × 200 = 80,000 bits
- 80,000 ÷ 8 = 10,000 bytes
- 10,000 ÷ 1,000 = 10 KB

Worked example: sound file size

A sound is recorded with a sample rate of 8,000 Hz and a bit depth of 16 bits, and lasts 30 seconds.

- 8,000 × 30 × 16 = 3,840,000 bits
- 3,840,000 ÷ 8 = 480,000 bytes
- 480,000 ÷ 1,000 = 480 KB

If the sample rate is given in kHz, multiply by 1,000 first: 8 kHz = 8,000 Hz.

Worked example: text file size

A text file has 2,000 characters, stored in ASCII using 8 bits per character.

- 8 × 2,000 = 16,000 bits
- 16,000 ÷ 8 = 2,000 bytes = 2 KB

If the same text used 16 bits per character, the file would be twice as big: 4 KB.

Binary numbers

Denary is the normal number system with ten digits (0 to 9). Binary uses just two digits (0 and 1). Each place in an 8-bit binary number has a value, doubling from right to left:

128	64	32	16	8	4	2	1

- The **most significant bit** (MSB) is the leftmost bit, with the biggest value (128 in 8 bits).
- The **least significant bit** (LSB) is the rightmost bit, with the smallest value (1).
- With 8 bits you can store denary 0 to 255, which is binary 00000000 to 11111111.
- Binary numbers can have anywhere from 1 to 8 bits. Adding zeros on the left does not change the value: 11010 is the same as 00011010.

Worked example: denary to binary (156)

Start at the left. If the place value fits into what is left, write 1 and take it away; if not, write 0.

- 128 fits into 156: write 1, 156 - 128 = 28 left.
- 64 does not fit into 28: 0. 32 does not fit: 0.
- 16 fits: 1, 28 - 16 = 12 left. 8 fits: 1, 12 - 8 = 4 left. 4 fits: 1, 0 left.
- 2: 0. 1: 0.

128	64	32	16	8	4	2	1
1	0	0	1	1	1	0	0

156 = 10011100. Check: 128 + 16 + 8 + 4 = 156.

Worked example: binary to denary (01011010 and 11010)

128	64	32	16	8	4	2	1
0	1	0	1	1	0	1	0

Add the place values with a 1 under them: 64 + 16 + 8 + 2 = 90.

For 11010, line it up from the right (00011010): 16 + 8 + 2 = 26.

Hexadecimal

Hexadecimal (hex) uses 16 digits: 0 to 9, then A to F for 10 to 15. Two hex digits can show any value from 00 to FF, which is denary 0 to 255 (the same as 8 bits). People use hex because it is shorter than binary and easier to read, and it converts to binary very easily.

Denary	0 to 9	10	11	12	13	14	15
Hex	0 to 9	A	B	C	D	E	F

In a 2-digit hex number, the left digit is worth 16s and the right digit is worth 1s.

Worked example: denary to hex (200) and hex to denary (3F)

- 200 to hex: 200 ÷ 16 = 12 remainder 8. 12 is C, so 200 = C8. Check: 12 × 16 = 192, 192 + 8 = 200.
- 3F to denary: 3 × 16 = 48, F = 15, 48 + 15 = 63.

Worked example: binary to hex (10110110) and hex to binary (9E)

Split the binary into two nibbles (groups of 4 bits). Each nibble uses place values 8, 4, 2, 1.

- 10110110 splits into 1011 and 0110. 1011 = 8 + 2 + 1 = 11 = B. 0110 = 4 + 2 = 6. So 10110110 = B6.
- 9E: 9 = 1001 and E = 14 = 1110. So 9E = 10011110.
- Binary with fewer than 8 bits: pad with zeros on the left first. 11010 becomes 0001 1010, which is 1A.

You can also go between denary and hex through binary: 200 = 11001000 = 1100 1000 = C8.

Binary addition

Add from right to left, carrying like normal addition. The four rules:

Sum	Write	Carry
0 + 0	0	0
0 + 1 or 1 + 0	1	0
1 + 1	0	1
1 + 1 + 1 (with a carry)	1	1

Worked example: 00110101 + 01001110 (53 + 78)

	128	64	32	16	8	4	2	1
	0	0	1	1	0	1	0	1
+	0	1	0	0	1	1	1	0
Carry	1	1	1	1	1			
=	**1**	**0**	**0**	**0**	**0**	**0**	**1**	**1**

Answer: 10000011. Check: 128 + 2 + 1 = 131, and 53 + 78 = 131.

Overflow

An overflow error happens when the result of a calculation is too big to fit in the number of bits available. With 8 bits the biggest number is 255. If an addition gives more than 255, there is a carry out of the leftmost column that needs a 9th bit. That bit is lost, so the stored answer is wrong.

Worked example: overflow with 11001000 + 01100100 (200 + 100)

	(9th bit)	128	64	32	16	8	4	2	1
		1	1	0	0	1	0	0	0
+		0	1	1	0	0	1	0	0
Carry	1	1							
=	**1**	**0**	**0**	**1**	**0**	**1**	**1**	**0**	**0**

The real answer is 300 (100101100), which needs 9 bits. Only 8 bits can be stored, so the 9th bit is lost and the computer keeps 00101100, which is 44. This is an overflow error.

Binary shifts

A binary shift moves every bit left or right by a set number of places. Empty places are filled with 0, and bits that fall off the end are lost.

- A **left shift** multiplies the number: 1 place × 2, 2 places × 4, 3 places × 8.
- A **right shift** divides the number: 1 place ÷ 2, 2 places ÷ 4, 3 places ÷ 8. If a 1 falls off the right-hand end, the answer is rounded down, so it loses accuracy.
- If a left shift pushes a 1 off the left-hand end, the answer is wrong (an overflow).

Worked example: shifting 00010110 (22)

- Left shift 2 places: 00010110 becomes 01011000 = 64 + 16 + 8 = 88. (22 × 4 = 88.)
- Right shift 1 place: 00010110 becomes 00001011 = 8 + 2 + 1 = 11. (22 ÷ 2 = 11.)
- Right shift 2 places: 00010110 becomes 00000101 = 5. (22 ÷ 4 = 5.5, but the 1 that fell off is lost, so the answer is 5.)
- Left shift 1 place of 11000000 (192) gives 10000000 (128), not 384, because the leftmost 1 is lost.

Key terms

Binary
A number system that uses only the digits 0 and 1.

Denary
The everyday number system that uses the digits 0 to 9.

Hexadecimal
A number system with 16 digits, 0 to 9 and A to F.

Bit
A single binary digit, 0 or 1.

Nibble
4 bits.

Byte
8 bits.

Kilobyte (KB)
1,000 bytes.

Megabyte (MB)
1,000 kilobytes.

Gigabyte (GB)
1,000 megabytes.

Terabyte (TB)
1,000 gigabytes.

Petabyte (PB)
1,000 terabytes.

Capacity
The fixed amount of data a storage device can hold.

Metadata
Extra data stored in a file that describes the file, such as an image's width and height.

Most significant bit (MSB)
The leftmost bit of a binary number, with the highest place value.

Least significant bit (LSB)
The rightmost bit of a binary number, with the place value 1.

Binary addition
Adding two binary numbers column by column from the right, carrying 1 when a column adds up to 2 or more.

Overflow error
An error that happens when a result is too big to fit in the number of bits available.

Binary shift
Moving all the bits in a binary number left (to multiply) or right (to divide) by a number of places.

Exam mistakes that lose marks

- **Forgetting units.** "480" on its own scores less than "480 KB". Always write the unit, and give the unit the question asks for.
- **Forgetting to divide by 8.** The formulas give bits. Divide by 8 to get bytes before dividing by 1,000.
- **Mixing 1,000 and 1,024 in one answer.** Either is accepted, but stick to one.
- **Using kHz as if it were Hz.** 44.1 kHz is 44,100 Hz. Convert first.
- **Using minutes instead of seconds for duration.** 2 minutes is 120 seconds.
- **Not showing working.** Without a calculator, a small slip can wipe out the final answer. Working can still get method marks.
- **Giving the wrong number of bits.** If the question asks for 8 bits, write all 8, including leading zeros (00011010, not 11010).
- **Losing the carry in binary addition.** Write carries clearly under or above the next column.
- **Explaining overflow as "the number is too big".** Say the result needs more bits than are available (more than 8), so the extra bit is lost and the stored answer is wrong.
- **Shifting the wrong way.** Left multiplies, right divides. Fill the gaps with 0.

- **Writing hex digits above 9 as numbers.** 12 is C, not "12". 1100 1000 is C8, not 128.

Quick check

1. Convert denary 99 to 8-bit binary.
2. Convert the hexadecimal number 2D to denary.
3. Add 10011001 and 01110011, and say whether an overflow has occurred.
4. Carry out a left shift of 3 places on 00000101. What has happened to the number?
5. An image is 100 pixels wide and 50 pixels high with a colour depth of 8 bits. Calculate its file size in KB.

▼ **Answers**

1. 99 = 64 + 32 + 2 + 1, so 01100011.
2. 2 × 16 = 32, D = 13, 32 + 13 = 45.
3. 10011001 (153) + 01110011 (115) = 268, which is more than 255. The 8-bit result is 00001100 (12) with a carry out into a 9th bit, so an overflow has occurred.
4. 00101000. The number has been multiplied by 8: 5 became 40.
5. 8 × 50 × 100 = 40,000 bits; 40,000 ÷ 8 = 5,000 bytes; 5,000 ÷ 1,000 = 5 KB.

1.2c Characters, images, sound and compression

Spec points 1.2.4 Data storage (characters, images, sound) and 1.2.5 Compression

What you need to know

Characters

Every character (letter, digit, punctuation mark, space or symbol) is given a unique binary code. When you press a key, the computer stores that character's code.

A **character set** is the complete list of characters a computer can use, with the binary code for each one.

The number of bits used per character limits how many characters a character set can hold. Each extra bit doubles the number of characters:

Bits per character	Number of characters
7	$2^7 = 128$
8	$2^8 = 256$
16	$2^{16} = 65{,}536$

ASCII and Unicode

	ASCII	Unicode
Bits per character	Originally 7 bits; extended ASCII uses 8 bits. In the exam, ASCII in binary will always be shown with 8 bits.	More bits per character (often 16, and up to 32)
Number of characters	128 (7 bits) or 256 (8 bits)	Many thousands; more than a million are possible
What it covers	English letters, digits, punctuation and some symbols	Characters for nearly every language in the world, plus symbols and emoji
Impact	Smaller files, but cannot show most other languages	Files take up more space, but text can be shared and shown correctly in any language

Unicode includes the ASCII characters with the same codes, so ASCII text still works in Unicode.

Character sets are logically ordered. Codes for letters run in alphabetical order, and digits run in number order. So if you know the code for 'A', the code for 'B' is one more, 'C' is two more, and so on. Upper case and lower case letters have different codes. You do not need to memorise any codes; the exam will give you one to work from.

> **Worked example: working out a character code**
>
> In ASCII, 'A' has the code 65 (01000001). What is the code for 'E'?
>
> E is 4 letters after A, so its code is 65 + 4 = 69. In 8-bit binary: 69 = 64 + 4 + 1 = 01000101.

Images

A bitmap image is made up of a grid of tiny dots called **pixels**. Each pixel is one colour, and that colour is stored as a binary code. The whole image is stored as the series of codes for every pixel.

- **Colour depth** is the number of bits used to store the colour of each pixel. The number of colours available is 2 to the power of the colour depth. 1 bit gives 2 colours (for example 0 = white, 1 = black). 2 bits gives 4 colours. 8 bits gives 256 colours. 24 bits gives over 16 million colours.
- **Resolution** is the number of pixels in the image, given as width × height (for example 1920 × 1080).
- **Metadata** is extra information stored in the image file, such as its width, height and colour depth. Without it the computer would not know how to rebuild the grid of pixels. Metadata can also include other details, such as the date the image was created.

Change	Effect on quality	Effect on file size
Increase colour depth	More colours available, so the image looks more realistic, with smoother changes of colour	Bigger, because each pixel needs more bits
Decrease colour depth	Fewer colours, so the image may look blocky or unrealistic	Smaller
Increase resolution	More pixels, so more detail and a sharper image	Bigger, because there are more pixels to store
Decrease resolution	Fewer pixels, so less detail; the image may look pixelated	Smaller

Image file size (bits) = colour depth × image height (px) × image width (px).

> **Worked example: changing colour depth**
>
> A 100 × 100 pixel image with a colour depth of 1 bit: 1 × 100 × 100 = 10,000 bits = 1,250 bytes. It can show only 2 colours.
>
> The same image with a colour depth of 8 bits: 8 × 100 × 100 = 80,000 bits = 10,000 bytes = 10 KB. It can show 256 colours, so it looks much better, but the file is 8 times bigger.

Sound

Sound in the real world is analogue: a smooth, continuous wave. Computers can only store binary, so sound has to be converted into digital form. This is done by **sampling**:

1. The height (amplitude) of the sound wave is measured at regular intervals. Each measurement is a sample.
2. Each sample is stored as a binary number.
3. When the sound is played back, the samples are used to rebuild the wave. It is close to the original, but not exactly the same.

- **Sample rate** is the number of samples taken each second, measured in hertz (Hz). Music is often sampled at 44,100 Hz (44.1 kHz).

- **Bit depth** is the number of bits available to store each sample (for example 16-bit).
- **Duration** is how many seconds of audio the sound file contains.

Change	Effect on playback quality	Effect on file size
Higher sample rate	Better: the samples are closer together, so the digital sound is closer to the original	Bigger: more samples to store each second
Higher bit depth	Better: each sample can be stored more precisely	Bigger: more bits for every sample
Longer duration	No change	Bigger: more seconds of samples to store
Lower sample rate or bit depth	Worse: the sound is less like the original	Smaller

Sound file size (bits) = sample rate (Hz) × duration (s) × bit depth.

> **Worked example: doubling the sample rate**
>
> A 10 second clip at 1,000 Hz and 8-bit: 1,000 × 10 × 8 = 80,000 bits = 10,000 bytes = 10 KB.
>
> At 2,000 Hz: 2,000 × 10 × 8 = 160,000 bits = 20 KB. Doubling the sample rate doubles the file size, and the quality improves.

The need for compression

Compression makes a file smaller. Common reasons it is needed:

- Files take up less storage space.
- Files are quicker to send over a network or the internet, and use less bandwidth.
- Email services limit the size of attachments, so large files may need compressing first.
- Music and video can be streamed without long waits.
- Web pages with compressed images load faster.

Lossy and lossless compression

	Lossy	Lossless
What happens to the data	Some data is permanently removed, usually detail people are unlikely to notice	No data is lost; the file is stored more efficiently
Can the original be restored?	No	Yes, exactly
File size	Much smaller	Smaller, but not as small as lossy
Quality	Lower than the original	Same as the original
Good for	Photos, music and video, especially for streaming and web pages (for example JPEG and MP3 files)	Text documents, program code and spreadsheets, where every bit must stay the same (for example PNG and ZIP files)
Not suitable for	Text or program files: losing data would change words or break the program	Situations where the smallest possible file is needed

You do not need to be able to carry out any specific compression method.

> **Worked example: choosing a compression type**
>
> - A band wants to stream songs to phones: lossy. The files become much smaller, so they stream quickly and use less data. Listeners are unlikely to notice the small loss of quality.
> - A student emails a program's source code: lossless. If any data were lost the program would not work, and the code must be restored exactly.

Key terms

Character
A single letter, digit, symbol, punctuation mark or space.

Character set
The complete list of characters a computer can use, with a unique binary code for each.

ASCII
A character set that uses 7 bits (128 characters), or 8 bits in extended ASCII (256 characters), mainly for English.

Unicode
A character set that uses more bits per character than ASCII so it can represent characters from nearly every language, plus symbols and emoji.

Pixel
The smallest single-colour dot that makes up a bitmap image.

Colour depth
The number of bits used to store the colour of each pixel.

Resolution

The number of pixels in an image, given as width × height.

Metadata
Extra information stored with a file, such as an image's width, height and colour depth.

Analogue
A continuous signal, such as a real sound wave, that is not made of separate values.

Digital
Data stored as separate binary values.

Sample
A measurement of the amplitude of a sound wave at one moment, stored as a binary number.

Sample rate
The number of samples taken per second, measured in hertz (Hz).

Bit depth
The number of bits available to store each sound sample.

Duration
How many seconds of audio a sound file contains.

Compression
Reducing the size of a file.

Lossy compression
Compression that permanently removes some data, so the original cannot be restored exactly.

Lossless compression
Compression that reduces file size without losing any data, so the original can be restored exactly.

Exam mistakes that lose marks

- **Saying Unicode "uses more memory" with nothing else.** Give the trade-off: more bits per character means more characters (many languages) but bigger files.
- **Thinking you must memorise ASCII codes.** You do not. Use the code you are given and count on through the alphabet.
- **Writing ASCII in 7 bits in the exam.** OCR will use 8 bits, so give 8-bit answers.
- **Saying higher resolution means "bigger pixels".** It means more pixels, which gives more detail.
- **Only saying "better quality" or "bigger file".** Explain why: more bits per pixel, more pixels, more samples per second or more bits per sample.
- **Saying a longer duration improves sound quality.** It only makes the file bigger.
- **Mixing up sample rate and bit depth.** Sample rate is how often samples are taken (Hz); bit depth is how many bits each sample uses.
- **Saying lossless compression "loses a little quality".** Lossless loses nothing; the original can be restored exactly.
- **Recommending lossy compression for text or program files.** Losing data would change the text or break the program.

- **Calling metadata "the image data".** Metadata is data about the image (height, width, colour depth), not the pixels themselves.

Quick check

1. Define the term "character set".
2. In a character set, 'a' has the code 97. State the code for 'd'.
3. Explain the effect of increasing the colour depth of an image.
4. Explain how analogue sound is stored in digital form.
5. Give one advantage and one disadvantage of lossy compression.

▼ **Answers**

1. The complete list of characters a computer can use, each with its own unique binary code.
2. 100 (d is 3 letters after a: 97 + 3).
3. More bits are used for each pixel, so more colours can be shown and the image looks more realistic, but the file size increases.
4. The amplitude of the sound wave is measured (sampled) at regular intervals, and each sample is stored as a binary number. The sample rate is how many samples are taken each second, and the bit depth is how many bits store each sample.
5. Advantage: much smaller files, so they take less space and are quicker to send or stream. Disadvantage: data is permanently removed, so quality is lower and the original cannot be restored.

1.3a Networks, topologies and the internet

Spec point 1.3.1 Networks and topologies

What you need to know

Types of network: LAN and WAN

	LAN (Local Area Network)	WAN (Wide Area Network)
Area covered	A small area, such as one building or one site	A large area: across towns, countries or the world
Who owns the hardware	Usually the organisation that uses it	Often rented from telecommunications companies that own the cables and links
Connection	Cables (such as Ethernet) or Wi-Fi	Telephone lines, fibre optic cables, satellite links; a WAN connects LANs together
Examples	A home network; a school network; an office network	The internet (the biggest WAN); a bank's network connecting all its branches

Factors that affect network performance

- **Bandwidth:** the amount of data that can be sent across the network in a given time, measured in bits per second. More bandwidth means data transfers faster.
- **Number of devices connected:** devices share the bandwidth. The more devices sending data at the same time, the more traffic and the slower the network for everyone.
- **Transmission media:** wired connections are usually faster and more reliable than wireless; fibre optic cable is faster than copper cable.
- **Interference and distance (wireless):** walls and other devices can weaken wireless signals, and the signal gets weaker the further you are from the access point.

Client-server and peer-to-peer networks

	Client-server	Peer-to-peer
Roles of the computers	Servers provide services (such as files, web pages, email or printing). Clients request and use those services.	All computers are equal. Each one can share its own files and resources directly with the others, with no central server.
Advantages	Files are stored centrally and can be backed up in one go; security, user accounts and software updates are managed from one place	Cheap and easy to set up; no expensive server; if one computer fails, the others keep working
Disadvantages	Servers and the staff to run them are expensive; if the server fails, clients lose access to its services	No central backup; files are spread across computers; each computer must look after its own security; a shared file is unavailable if its computer is off
Typical use	Schools, businesses	Small home networks; sharing files directly between devices

Servers provide different services. A **web server** stores web pages and sends them to clients that ask for them. A **file server** stores files and lets clients save and retrieve them.

Hardware needed to build a LAN

Hardware	What it does
NIC (Network Interface Controller/Card)	Lets a device connect to a network, by cable or wirelessly. Every NIC has a unique MAC address.
Switch	Connects devices together on a LAN. It receives data and sends it only to the device it is meant for, using its MAC address.
Router	Sends data between different networks, for example between a LAN and the internet. It uses IP addresses to forward packets of data towards their destination.
Wireless access point (WAP)	Lets wireless devices connect to a wired network using Wi-Fi.
Transmission media	What carries the data: copper cables (such as Ethernet cable), fibre optic cables, or radio waves for wireless.

At home, one box (often called a router) usually contains a router, a switch and a wireless access point together.

The internet

The internet is a worldwide collection of computer networks: a network of networks, all connected together.

DNS. Computers on the internet are found by their IP address, but people use web addresses (URLs) such as www.example.com because they are easier to remember. The Domain Name Service (DNS) converts URLs into IP addresses. DNS is made up of many Domain Name Servers around the world, each holding a list of domain names and their IP addresses.

1. You type a URL into a web browser.
2. The browser sends a request to a Domain Name Server to look up the IP address for that domain name.
3. If that server has the domain name in its records, it sends back the IP address. If not, it passes the request on to another Domain Name Server until the address is found.
4. The IP address is returned to the browser.
5. The browser uses the IP address to contact the web server and request the web page.

Hosting. Hosting means storing a website or files on a server that is connected to the internet, so other people can access them. Companies called hosts rent out space on their servers.

Web servers and clients. A web server stores web pages and sends them out when they are requested. The client (your web browser) sends a request for a page, receives it and displays it.

The Cloud. The Cloud means services provided remotely over the internet, on servers owned by another company. Examples are cloud storage (saving files online), software (running apps in a browser) and processing (using another company's computers to do the work).

Advantages of the Cloud	Disadvantages of the Cloud
• Access files and software from any device, anywhere with an internet connection • Easy to share and work on files with other people • No need to buy and look after your own hardware • Easy to get more storage when you need it • The provider handles backups, security and updates	• Needs an internet connection; no connection means no access • Slow connections make it slow to use • Your data is held by another company, which raises security and privacy worries • Usually an ongoing subscription cost • If the provider has a problem or closes, you may lose access

Star and Mesh topologies

A topology is the way the devices in a network are connected together.

Star: every device is connected by its own cable to a central switch.

Mesh: devices are connected to each other with many links, so data can travel along different routes. In a full mesh, every device connects to every other device. In a partial mesh, devices connect to some of the others. Mesh networks are often wireless, which avoids needing lots of cables.

	Star	Mesh
Advantages	• If one cable or device fails, only that device is affected • Easy to add new devices • Fast and reliable, because the switch sends data only to the device it is meant for	• No single point of failure: if one link or device fails, data takes another route • Very reliable and robust • Data can travel by the quickest available route
Disadvantages	• If the central switch fails, the whole network goes down • Needs a lot of cable, because every device has its own cable to the switch	• A wired full mesh needs a huge amount of cable, so it is expensive • More complicated to set up and manage

Worked example: choosing a topology

A hospital needs a network that must keep working even if a connection breaks. A small office of eight computers wants a network that is cheap and easy to manage.

- Hospital: mesh. If one link fails, data is sent another way, so there is no single point of failure. The extra cost is worth it because the network must not go down.
- Office: star. It is cheap and simple, new computers are easy to add, and one faulty cable only affects one computer. The risk is the central switch, so the office could keep a spare.

Key terms

Network
Two or more computers or devices connected together so they can share data and resources.

Stand-alone computer
A computer that is not connected to a network.

LAN (Local Area Network)
A network that covers a small area, such as one building or site, with hardware usually owned by the organisation.

WAN (Wide Area Network)
A network that covers a large geographical area and connects LANs together, often using rented infrastructure.

Bandwidth
The amount of data that can be sent across a network in a given time, measured in bits per second.

Client
A computer that requests and uses services from a server.

Server
A computer that provides services, such as files, web pages or email, to clients.

Client-server network

A network where clients request services from one or more central servers.

Peer-to-peer network
A network where all computers are equal and share resources directly with each other.

NIC (Network Interface Controller/Card)
Hardware that lets a device connect to a network.

Switch
A device that connects devices on a LAN and sends data only to the device it is meant for, using MAC addresses.

Router
A device that sends data between networks, such as a LAN and the internet, using IP addresses.

Wireless access point (WAP)
A device that lets wireless devices connect to a wired network.

Transmission media
The cables or radio waves that carry data across a network.

Internet
A worldwide collection of computer networks.

URL
A web address, such as www.example.com, that people use to find a website.

DNS (Domain Name Service)
The system, made up of many Domain Name Servers, that converts URLs into IP addresses.

Domain Name Server
A server that holds domain names and their matching IP addresses and answers look-up requests.

Hosting
Storing a website or files on a server connected to the internet so others can access them.

Web server
A server that stores web pages and sends them to clients that request them.

The Cloud
Services such as storage, software and processing provided remotely over the internet.

Topology
The way devices on a network are connected together.

Star topology
A topology where every device connects to a central switch.

Mesh topology
A topology where devices are connected to many others, so data can take more than one route.

Exam mistakes that lose marks

- **Saying a WAN is just "a big LAN".** Say it covers a large geographical area, connects LANs together and often uses infrastructure rented from telecommunications companies.

- **Defining a LAN by the number of computers.** A LAN is defined by the small area it covers, not how many devices it has.
- **Mixing up a switch and a router.** A switch connects devices within one LAN (using MAC addresses); a router connects different networks (using IP addresses).
- **Saying a wireless access point gives you internet access.** It connects wireless devices to a network. Reaching the internet needs a router.
- **Calling the internet and the World Wide Web the same thing.** The internet is the network of networks; web pages are one service that uses it.
- **Saying DNS "finds the website".** Say it converts the URL into an IP address, which the browser then uses to contact the web server.
- **Saying the Cloud means data is stored "nowhere" or "in the sky".** It is stored on servers owned by another company, accessed over the internet.
- **Saying a star network fails if any device fails.** Only the failure of the central switch brings the whole network down.
- **Giving advantages of a topology without linking to the scenario.** Say why it suits that organisation.

Quick check

1. Give two differences between a LAN and a WAN.
2. State two factors that can affect the performance of a network.
3. Describe the role of a router.
4. Describe how DNS is used when you type a URL into a browser.
5. Give one advantage and one disadvantage of a mesh topology.

▼ **Answers**

1. Any two: a LAN covers a small area, a WAN a large area; LAN hardware is usually owned by the organisation, WAN infrastructure is often rented; a WAN connects LANs together.
2. Any two: bandwidth; number of devices connected; type of transmission media; interference or distance for wireless.
3. A router sends data between different networks, such as a LAN and the internet, using IP addresses to forward packets towards their destination.
4. The browser sends the URL to a Domain Name Server, which looks up the matching IP address (passing the request to other DNS servers if it does not have it) and returns it. The browser then uses the IP address to contact the web server.
5. Advantage: no single point of failure, as data can take another route if a link fails. Disadvantage: a wired mesh needs a lot of cable, so it is expensive and complicated to set up.

1.3b Wired and wireless networks, protocols and layers

Spec point 1.3.2 Wired and wireless networks, protocols and layers

What you need to know

Modes of connection

Wired: Ethernet. Devices are connected with cables (copper Ethernet cable or fibre optic cable).

Wireless: Wi-Fi. Devices connect to a wireless access point using radio waves, usually within a building.

Wireless: Bluetooth. Devices connect directly to each other using radio waves over a short distance (usually around 10 metres), for example wireless headphones, a phone and a car, or a wireless keyboard. It uses little power but has a low bandwidth.

	Wired (Ethernet)	Wireless (Wi-Fi)
Speed	Usually faster	Usually slower, and drops further from the access point
Reliability	Stable connection, little interference	Signals weakened by walls, distance and interference from other devices
Security	More secure: someone needs physical access to a cable	Less secure: signals travel through the air and can be intercepted, so encryption is needed
Mobility	Devices are tied to where the cable is	Users can move around freely
Setting up	Cables cost money to buy and fit, and can be a trip hazard	Easy to add new devices; no cables needed

You do not need to know how the Ethernet, Wi-Fi and Bluetooth protocols work.

> **Worked example: recommending connections**
>
> A small design company has desktop computers that send large video files, and staff who use laptops and phones in meetings.
>
> - Desktops: wired Ethernet. The large files transfer faster over cable, the connection is stable, and the desktops do not move.
> - Laptops and phones: Wi-Fi. Staff can move between rooms and meetings without cables, and new devices are easy to connect. The Wi-Fi should be encrypted to stop data being intercepted.
> - Wireless headsets for video calls: Bluetooth. Short range and low power is all a headset needs.

Encryption

Encryption scrambles data so that anyone who intercepts it cannot understand it. The original data (plaintext) is turned into scrambled data (ciphertext) using an encryption key. Only someone with the correct key can decrypt it back to the original. Encryption does not stop data

being intercepted; it makes intercepted data useless. It is especially important on wireless networks, where signals can be picked up by anyone in range.

IP addressing

An IP address identifies a device on a network so that data can be sent to it across networks. Routers use IP addresses to send data to the right place.

- **IPv4:** 32 bits, written as four denary numbers from 0 to 255 separated by full stops, for example 192.168.1.10.
- **IPv6:** 128 bits, written as eight groups of four hexadecimal digits separated by colons, for example 2001:0db8:85a3:0000:0000:8a2e:0370:7334. IPv6 was introduced because the world was running out of IPv4 addresses.

You do not need to know the differences between static and dynamic, or public and private, IP addresses.

MAC addressing

A MAC address is a unique address given to a device's NIC by the manufacturer. It is fixed and does not change. It is used within a network (a LAN): a switch uses MAC addresses to send data to the exact device it is meant for.

Format: 48 bits, written as six pairs of hexadecimal digits separated by colons (or hyphens), for example 3C:5A:B4:12:9F:0E.

	IP address	**MAC address**
Identifies	A device's location on a network, for sending data across networks	A device's NIC, for sending data within a LAN
Can it change?	Yes, depending on the network the device joins	No, it is fixed by the manufacturer
Format	IPv4: four denary numbers 0 to 255; IPv6: eight groups of hex	Six pairs of hex digits

Standards

A standard is an agreed set of rules for an area of computing, such as how hardware connects or how a file format is laid out. Standards let hardware and software made by different manufacturers work together. For example, any device that follows the Wi-Fi standards can connect to any Wi-Fi network, whoever made it.

You do not need to know any individual standards.

Protocols

A protocol is a set of rules for transferring data between devices. Different protocols are used for different purposes.

Protocol	Purpose and key features
TCP/IP (Transmission Control Protocol/Internet Protocol)	The main set of rules for sending data across the internet. TCP splits data into packets, numbers them, checks they all arrive, asks for any missing ones to be sent again, and puts them back in order. IP adds the IP addresses and routes each packet to the right destination.
HTTP (Hyper Text Transfer Protocol)	Used by web browsers to request web pages from web servers, and by servers to send them back.
HTTPS (Hyper Text Transfer Protocol Secure)	The same as HTTP, but the data is encrypted, so it cannot be read if intercepted. Used for logins, online banking and shopping.
FTP (File Transfer Protocol)	Used to transfer files between computers over a network, for example uploading website files to a web server.
POP (Post Office Protocol)	Used to retrieve email from a mail server. The emails are downloaded to one device and are usually deleted from the server.
IMAP (Internet Message Access Protocol)	Used to retrieve email from a mail server. The emails stay on the server, so they can be read on several devices and stay in sync.
SMTP (Simple Mail Transfer Protocol)	Used to send email from a device to a mail server, and between mail servers.

The concept of layers

Network protocols are grouped into layers. Each layer has its own specific job and only deals with the layer directly above it and the layer directly below it. As data is sent, it passes down through the layers; when it arrives, it passes back up through them.

Benefits of using layers:

- Each layer is self-contained, so one layer can be changed or improved without affecting the others.
- A big, complicated problem is broken down into smaller parts that are easier to design and fix.
- Developers and manufacturers can work on one layer without needing to know how the other layers work.
- Layers follow standards, so hardware and software from different companies can work together.

OCR uses the 4-layer TCP/IP model (application, transport, internet and link layers) as an example. You do not need to learn the names or jobs of each layer.

Key terms

Ethernet
: A wired way of connecting devices on a network using cables.

Wi-Fi
: A wireless way of connecting devices to a network using radio waves.

Bluetooth
A short-range wireless way of connecting devices directly to each other.

Encryption
Scrambling data with a key so that it cannot be understood by anyone without the key.

Encryption key
The value used to encrypt and decrypt data.

IP address
An address that identifies a device on a network so data can be routed to it.

IPv4
A 32-bit IP address written as four denary numbers from 0 to 255 separated by full stops.

IPv6
A 128-bit IP address written as eight groups of four hexadecimal digits separated by colons.

MAC address
A unique, fixed 48-bit address given to a NIC by its manufacturer, written as six pairs of hex digits.

Standard
An agreed set of rules for an area of computing that lets products from different manufacturers work together.

Protocol
A set of rules for transferring data between devices.

Packet
A small piece of data sent across a network, with the addresses needed to deliver it.

TCP/IP
Protocols that split data into packets, route them across networks and put them back together in order.

HTTP
The protocol used to request and send web pages.

HTTPS
A secure version of HTTP that encrypts the data being sent.

FTP
The protocol used to transfer files between computers over a network.

POP
An email protocol that downloads emails from the server to one device, usually deleting them from the server.

IMAP
An email protocol that keeps emails on the server so they can be accessed from several devices.

SMTP
The protocol used to send emails.

Layer
A group of protocols with one specific job, which only deals with the layers directly above and below it.

Exam mistakes that lose marks

- **Saying wireless is "always worse".** Compare properly: wired is usually faster and more secure; wireless gives mobility and is easier to set up.
- **Saying encryption stops data being intercepted.** It does not stop interception; it makes intercepted data unreadable without the key.
- **Mixing up IP and MAC addresses.** IP addresses are for routing data across networks and can change; MAC addresses are fixed, belong to the NIC and are used within a LAN.
- **Getting the formats wrong.** IPv4 is four denary numbers (each 0 to 255) with full stops. IPv6 and MAC addresses use hexadecimal.
- **Mixing up POP, IMAP and SMTP.** SMTP sends email. POP and IMAP both receive it: POP downloads and usually deletes from the server; IMAP keeps it on the server.
- **Saying HTTPS is "faster" than HTTP.** The difference is that HTTPS encrypts the data.
- **Confusing a standard with a protocol.** A protocol is a set of rules for transferring data; a standard is a wider agreed set of rules that lets different manufacturers' products work together.
- **Explaining layers by listing the TCP/IP layer names.** You do not need them. Explain the benefits: each layer can be changed on its own, and problems are broken into smaller parts.

Quick check

1. Give two benefits of a wired connection compared with a wireless one.
2. State the format of an IPv4 address.
3. Explain why a protocol is needed.
4. Which protocol is used to send email, and which keeps received emails on the server?
5. Give two benefits of using layers in networking.

▼ **Answers**

1. Any two: usually faster; more reliable with less interference; more secure because you need physical access to a cable.
2. 32 bits, written as four denary numbers from 0 to 255 separated by full stops (for example 192.168.1.10).
3. A protocol sets the rules for transferring data, so that devices (even from different manufacturers) send and receive data in a way both understand.
4. SMTP sends email. IMAP keeps received emails on the server.
5. Any two: one layer can be changed without affecting the others; a complex problem is split into smaller parts that are easier to design and fix; developers can work on one layer without knowing how the others work; layers follow standards so products from different companies work together.

1.4 Network security

Spec points 1.4.1 Threats to computer systems and networks and 1.4.2 Identifying and preventing vulnerabilities

What you need to know

Forms of attack

For each attack, learn how it is used and what its purpose is.

Malware is malicious software: any software written to harm a computer or its user.

- **Virus:** attaches itself to a file or program and spreads when that file is run or shared. It can delete or damage files.
- **Worm:** copies itself and spreads across a network on its own, without the user doing anything. It can slow networks down by using up bandwidth.
- **Trojan:** pretends to be useful, legitimate software, so the user installs it. It then harms the computer or lets an attacker in.
- **Ransomware:** encrypts the user's files and demands payment to unlock them.
- **Spyware:** secretly records what the user does, such as the keys they press, and sends it to the attacker to steal passwords and bank details.

Purpose: to damage or delete files, steal data, take control of a computer, or make money by blackmail.

Social engineering means tricking people into giving away information or access, rather than attacking the computer itself. People are often the weak point in a system: they can be fooled even when the technology is secure.

- **Phishing:** fake emails, texts or websites that pretend to be from a trusted organisation, such as a bank, to get people to enter passwords or bank details, or to click a link that installs malware.
- Other examples: phoning someone and pretending to be from the IT department to get their password; watching over someone's shoulder as they type a PIN.

Purpose: to get passwords, personal or financial details, or access to a system.

Brute-force attack: software tries every possible password (or a huge list of common ones) one after another until it finds the right one. Purpose: to get into an account or system without permission.

Denial of service (DoS) attack: floods a server or network with so many requests that it cannot cope, so real users cannot use it. When many computers are used at once, it is a distributed denial of service (DDoS) attack. Purpose: to take a website or service offline, cost a business money, or make a protest.

Data interception and theft: data is captured as it travels across a network (for example with software that reads packets on the network), or is stolen from a device or storage. Purpose: to steal passwords, personal data or financial information.

SQL injection: SQL is the language used to work with databases. In an SQL injection attack, the attacker types SQL code into an input box on a website, such as a login form. If the website does not check the input, the database runs the code. Purpose: to see, change or delete data in the database, or get past a login without a password.

Prevention methods

Method	How it works	What it limits or prevents
Penetration testing	An organisation pays authorised people to try to attack its own system, the way a criminal would, to find weaknesses. The weaknesses are then fixed.	Finds vulnerabilities before criminals do, so any type of attack is less likely to succeed.
Anti-malware software	Scans files and compares them against a database of known malware. Anything found is quarantined or deleted. It must be updated regularly to recognise new malware.	Malware, such as viruses, worms, trojans and spyware.
Firewall	Hardware or software that checks all data coming into and going out of a network or computer, and blocks anything that breaks its set of rules (for example traffic from a suspicious address).	Unauthorised access from outside, some malware arriving over the network, and some denial of service traffic.
User access levels	Each user is given access only to the files and functions they need for their job (for example read-only, or full access for an administrator).	Limits the damage if an account is misused or stolen, and stops staff seeing data they should not.
Passwords	Users must enter a password to log in. Strong passwords are long and use a mix of upper and lower case letters, numbers and symbols. Accounts can lock after a few wrong attempts.	Unauthorised access. Strong passwords and lockouts make brute-force attacks much harder.
Encryption	Scrambles data with a key so it is unreadable without the key.	Makes intercepted or stolen data useless to the attacker.
Physical security	Locks on server rooms, security guards, CCTV, ID cards, keypads and biometric locks (such as fingerprint scanners).	People getting into rooms to steal or damage hardware, or plugging in devices to steal data.

Worked example: matching threats to prevention

Threat	Best ways to limit it
Malware	Anti-malware software (kept up to date), firewall, training staff not to open unknown attachments
Phishing and other social engineering	Training staff to spot fake messages, user access levels to limit what a tricked user can reach
Brute-force attack	Strong passwords, locking accounts after several failed attempts
Denial of service	Firewall to block suspicious traffic
Data interception and theft	Encryption, physical security, user access levels
SQL injection	Checking (validating) all input on web forms before it reaches the database, user access levels on the database, penetration testing

Key terms

Malware
Malicious software written to harm a computer or its user.

Virus
Malware that attaches to a file and spreads when the file is run or shared.

Worm
Malware that copies itself and spreads across a network without user action.

Trojan
Malware disguised as legitimate software.

Ransomware
Malware that encrypts a user's files and demands payment to unlock them.

Spyware
Malware that secretly records what a user does and sends it to an attacker.

Social engineering
Tricking people into giving away information or access.

Phishing
Fake emails, texts or websites that pretend to be from a trusted organisation to steal personal details.

Brute-force attack
Trying every possible password until the right one is found.

Denial of service (DoS) attack
Flooding a server or network with requests so real users cannot use it.

Data interception and theft
Capturing data as it travels across a network, or stealing it from a device.

SQL injection

Typing SQL code into a website's input box so the database runs it.

Vulnerability
A weakness in a system that an attacker could use.

Penetration testing
Authorised, simulated attacks on a system to find weaknesses so they can be fixed.

Anti-malware software
Software that finds and removes malware by comparing files with known malware.

Firewall
Hardware or software that checks incoming and outgoing network traffic and blocks anything that breaks its rules.

User access levels
Settings that control which files and functions each user can use.

Password
A secret string of characters a user enters to prove who they are.

Encryption
Scrambling data with a key so it cannot be read without the key.

Physical security
Measures such as locks, guards and CCTV that stop people physically reaching hardware.

Exam mistakes that lose marks

- **Saying a firewall stops all viruses.** A firewall filters network traffic. It does not scan files or remove malware already on a computer; that is the job of anti-malware software.
- **Saying encryption stops hackers getting the data.** Encryption does not stop interception or theft; it makes the data unreadable without the key.
- **Saying anti-malware protects against everything.** It only knows malware in its database, so it must be updated, and it does nothing against phishing calls or physical theft.
- **Describing a DoS attack as "hacking into" a server or stealing data.** It overloads the server so real users cannot get in.
- **Describing a brute-force attack as "guessing the password".** Say it tries every possible combination, using software, until it finds the right one.
- **Calling penetration testing an attack by criminals.** It is authorised testing, done to find and fix weaknesses.
- **Naming a prevention method without explaining it.** Say how it limits the attack: "accounts lock after three wrong attempts, so a brute-force attack cannot try every password".
- **Forgetting the human side.** For social engineering, staff training and user access levels are the key answers, because technology alone cannot stop people being tricked.

Quick check

1. Describe what a phishing attack is.
2. Explain why people are called the "weak point" in a system.

3. State the purpose of a denial of service attack.
4. Describe how a firewall protects a network.
5. Explain how strong passwords and account lockouts help against brute-force attacks.

▼ **Answers**

1. Fake emails, texts or websites that pretend to be from a trusted organisation, to trick people into giving away details such as passwords or bank details.
2. People can be tricked into giving away information or access even when the technology is secure, for example by clicking a phishing link or telling a caller their password.
3. To stop real users getting to a website or service by flooding it with requests, taking it offline.
4. It checks all data coming in and going out against a set of rules and blocks anything that breaks the rules, such as traffic from a suspicious address.
5. A strong password has so many possible combinations that trying them all would take far too long, and locking the account after a few wrong attempts stops the software trying more.

1.5 Systems software

Spec points 1.5.1 Operating systems and 1.5.2 Utility software

What you need to know

The purpose of an operating system

An operating system (OS) is software that manages the computer's hardware and software, and lets the user interact with the computer. Its main functions are below.

User interface

The user interface lets the user communicate with the computer: give it commands and see the results.

- **Graphical user interface (GUI):** uses windows, icons, menus and a pointer (WIMP), or touch on phones and tablets. Easy for beginners, as you do not need to remember commands, but it uses more memory and processing power.
- **Command-line interface:** the user types text commands. Uses very little memory and processing power and is fast for expert users, but you have to learn the commands.
- Other types include menu-based interfaces (such as on a cash machine) and voice interfaces.

Memory management and multitasking

- **Transfer of data between memory:** the OS moves programs and data from secondary storage into RAM when they are needed, and back out when they are not. When RAM is full, it uses virtual memory.
- **Allocation of memory to applications:** the OS gives each running program its own area of RAM, makes sure programs do not overwrite each other's memory, and frees up the memory when a program is closed.
- **Multitasking:** because several programs can be held in memory at once, the OS can let them all run "at the same time". It does this by switching the CPU between them very quickly, so the user sees them all running together.

You do not need to know about paging or segmentation.

Peripheral management and drivers

Peripherals are devices connected to the computer, such as printers, keyboards, mice and monitors. Data is transferred between these devices and the processor, and this process needs to be managed. The OS manages it, for example by controlling the flow of data to a printer so the processor does not have to wait.

A **driver** is a small piece of software that lets the OS communicate with a particular device. It translates the OS's general instructions into commands that specific device understands. A new device needs the right driver installed before it will work, and drivers are updated to fix problems.

User management

- **Allocation of an account:** creating, changing and deleting user accounts, each with a username and password.
- **Access rights:** controlling which files, folders and programs each user can use (for example a student cannot open teachers' files).

- **Security:** checking passwords at login, locking accounts after too many wrong attempts, and keeping each user's files private from other users.

File management

The OS organises the files stored in secondary storage. Its key features include:

- **Naming** and renaming files.
- **Allocating to folders:** organising files into folders and sub-folders (a hierarchy).
- **Moving**, copying and deleting files.
- **Saving** and opening files, and keeping track of where each file is stored.
- Setting who is allowed to read or change each file.

Utility software

Computers often come with utility software. Utility software is system software that performs housekeeping tasks: jobs that help keep the computer running well and its data safe. It is needed for additional tasks that the operating system may not carry out on its own.

Utility	Purpose	Why it is needed
Encryption software	Scrambles files using a key so they cannot be read without the key.	Keeps data secure if a device is lost or stolen, or if files are intercepted while being sent.
Defragmentation	On a magnetic hard disk drive, files get split into pieces (fragments) stored in different places as files are saved, changed and deleted. Defragmentation moves the pieces so each file is stored together, and gathers the free space together.	The drive no longer has to jump between different places to read one file, so files open faster. It is not needed for solid state drives, which have no moving parts and read all locations equally fast.
Data compression	Reduces the size of files (for example by putting them into a zip file).	Files take up less storage space and are quicker to send over a network or attach to an email.

> **Worked example: why defragmentation speeds up a hard drive**
>
> A 10 MB file is saved on a nearly full hard drive. There is no single gap big enough, so it is stored in three fragments in different places. To open it, the drive has to move to three locations, which takes time. After defragmentation, the three fragments are stored next to each other, so the drive reads the file in one go and it opens faster.

Key terms

System software
Software that runs and manages the computer, such as the operating system and utilities.

Operating system (OS)
Software that manages a computer's hardware and software and provides a user interface.

User interface

The part of the OS that lets the user give commands and see results.

Graphical user interface (GUI)
An interface that uses windows, icons, menus and a pointer, or touch.

Command-line interface
An interface where the user types text commands.

Memory management
The OS moving data between storage and RAM and giving each program its own area of memory.

Multitasking
Running more than one program at the same time by switching the CPU quickly between them.

Peripheral
A device connected to a computer, such as a printer, keyboard or monitor.

Peripheral management
The OS managing the transfer of data between devices and the processor.

Driver
Software that lets the OS communicate with a particular hardware device.

User management
The OS creating user accounts, setting access rights and keeping accounts secure.

Access rights
Permissions that control which files and programs a user can use.

File management
The OS naming, saving, moving and organising files into folders.

Utility software
System software that carries out housekeeping tasks to keep the computer running well and its data safe.

Encryption software
A utility that scrambles files so they cannot be read without a key.

Defragmentation
A utility that rearranges fragmented files on a hard disk drive so each file is stored together.

Fragmentation
When a file is split into pieces stored in different places on a disk.

Data compression software
A utility that reduces the size of files.

Exam mistakes that lose marks

- **Giving vague answers such as "the OS runs the computer".** Name the function and say what it does: "memory management allocates RAM to each program so they do not overwrite each other".
- **Thinking multitasking means the CPU runs programs at exactly the same time.** On one core, the OS switches between programs so quickly they appear to run together.
- **Saying a driver is hardware.** A driver is software that lets the OS talk to a device.

- **Confusing user management with file management.** User management is about accounts and access rights; file management is about naming, saving, moving and organising files.
- **Saying defragmentation frees up space or deletes files.** It rearranges file fragments so they are stored together; the amount of free space stays the same.
- **Recommending defragmentation for an SSD.** SSDs have no moving parts, so defragmentation gives no speed benefit.
- **Saying utility software is application software.** Utilities are system software that look after the computer.
- **Mixing up compression and encryption.** Compression makes files smaller; encryption makes them unreadable without a key.

Quick check

1. State three functions of an operating system.
2. Describe how memory management allows multitasking.
3. Explain what a device driver does.
4. Give two features of file management.
5. Explain why defragmentation can make a hard disk drive faster.

▼ **Answers**

1. Any three: user interface; memory management and multitasking; peripheral management and drivers; user management; file management.
2. The OS loads several programs into RAM at once, gives each its own area of memory so they do not overwrite each other, and switches the CPU between them quickly so they appear to run at the same time.
3. It is software that translates the OS's instructions into commands a particular device understands, so the OS can communicate with the device.
4. Any two: naming files; allocating files to folders; moving files; saving files; copying or deleting files; setting file permissions.
5. Over time, files are split into fragments stored in different places. Defragmentation moves the fragments so each file is stored together, so the drive does not have to move between places to read a file, and files load faster.

1.6 Ethical, legal, cultural and environmental impacts

Spec point 1.6.1 Ethical, legal, cultural and environmental impact

What you need to know

Digital technology changes how people live, work and communicate. This brings ethical, legal, cultural, environmental and privacy issues. In the exam you need to discuss these issues using examples, and know the purpose of the laws listed below and what each allows or prohibits. Practise giving points on more than one side.

The five types of issue

Issue	What it is about	Examples
Ethical	Whether something is right or wrong, and fair to everyone	Jobs replaced by automation and robots; who is to blame if a self-driving car crashes; companies using personal data to target adverts; the digital divide (people who cannot afford devices or internet access fall behind); cyberbullying
Legal	What the law allows or prohibits	Hacking; spreading malware; copying software, music or films illegally; misuse of personal data; laws differing between countries, so online crimes are hard to deal with
Cultural	How technology changes the way people live, behave and relate to each other	Social media changing how people communicate; working and learning from home; online shopping replacing high street shops; streaming replacing television schedules; a gap between people who use technology and people who do not
Environmental	The effect on the planet	E-waste: old devices thrown away contain toxic materials and often end up in landfill or shipped abroad; mining rare metals for devices; data centres using huge amounts of energy; on the plus side, video calls cut travel, digital documents save paper and smart systems can save energy
Privacy	People's right to keep their personal information to themselves	Location tracking on phones; CCTV and facial recognition; websites recording browsing habits; social media posts staying online; data breaches exposing personal details

Worked example: discussing one technology (self-driving cars)

- Ethical: could reduce accidents caused by human error, but it is unclear who is responsible when one crashes, and taxi and lorry drivers could lose their jobs.
- Legal: laws must decide who is liable in a crash: the owner, the manufacturer or the programmer.
- Cultural: people who cannot drive, such as some older or disabled people, could travel independently.
- Environmental: smoother driving could save fuel, but making the sensors and computers uses resources and energy.
- Privacy: the car records where you go and when, and that data could be shared or stolen.
- Conclusion: give a judgement, for example "the safety benefits are large, but clear laws on responsibility and data are needed first".

Legislation

The Data Protection Act 2018

Purpose: to control how organisations collect, store and use personal data, and to protect people's rights over their data. Organisations must make sure personal data is:

- used fairly, lawfully and openly (people are told what it is used for);
- used only for the purpose it was collected for;
- relevant and limited to what is needed;
- accurate and kept up to date;
- kept no longer than necessary;
- kept secure, protected from loss, theft and unauthorised access.

It gives people rights, including the right to see the data an organisation holds about them, to have wrong data corrected and, in some cases, to have their data deleted. It prohibits organisations from, for example, selling or sharing personal data for another purpose without permission, or keeping it insecurely.

Computer Misuse Act 1990

Purpose: to make hacking and other misuse of computers illegal. It created three main offences:

1. **Unauthorised access to computer material:** getting into a computer system or files without permission, for example logging into someone else's account using their password.
2. **Unauthorised access with intent to commit or facilitate further offences:** getting in without permission in order to commit another crime, for example hacking a bank's system to steal money.
3. **Unauthorised acts with intent to impair the operation of a computer:** deliberately damaging or disrupting a computer system or data, for example spreading malware, deleting files or carrying out a denial of service attack.

Copyright Designs and Patents Act 1988

Purpose: to protect people's original work and inventions from being copied or used without permission. Copyright covers things like software, music, films, books and images, and applies

automatically when the work is created. Patents protect new inventions. It makes it illegal to:

- copy, share or sell software, music, films or images without permission (for example piracy);
- use someone else's work and claim it as your own;
- install software on more computers than the licence allows.

Software licences

A software licence is a legal agreement that sets out how software can be used: for example how many devices it can be installed on, and whether it can be copied, changed or shared. Software needs a licence so the developer's work is legally protected and they can control how it is used and get paid for it.

	Open source	Proprietary
Features	The source code is available. Users can see it, change the software and share their changes. Usually free.	No access to the source code; users cannot change it. Usually paid for, commonly bought off-the-shelf or by subscription.
Benefits	Free to use; can be changed to suit your needs; a community of users can find and fix problems	Professionally developed and tested; official support and help; regular updates; the company can earn money from it
Drawbacks	May have little official support; quality and updates depend on volunteers; changes made by others may contain errors; you may need technical skill to change it	Costs money, often for each user or each year; cannot be adapted to your needs; licence limits how many devices can use it

Worked example: recommending a licence

Scenario 1: a small charity with no money needs office software, and a volunteer is a skilled programmer. Recommend open source: it is free, and the volunteer can change it to suit the charity. Drawback: there is no official support line if something goes wrong.

Scenario 2: a company is selling a game it spent two years writing. Recommend a proprietary licence: customers cannot see or copy the source code, so the company protects its work and can charge for it. Drawback: users cannot fix bugs themselves, so the company must provide support and updates.

Key terms

Ethical issue
A question about whether a use of technology is right, wrong or fair.

Legal issue
A question about what the law allows or prohibits.

Cultural issue
How technology changes the way people live, behave and interact.

Environmental issue

The effect of technology on the planet, such as energy use and e-waste.

Privacy issue
A question about people's right to control their personal information.

Digital divide
The gap between people who have access to digital technology and those who do not.

E-waste
Electronic devices that have been thrown away.

Personal data
Information about a living person who can be identified from it.

The Data Protection Act 2018
The law that controls how organisations collect, store and use personal data.

Computer Misuse Act 1990
The law that makes unauthorised access to, and misuse of, computer systems illegal.

Copyright Designs and Patents Act 1988
The law that protects original work and inventions from being copied or used without permission.

Copyright
The legal right of a creator to control how their original work is copied and used.

Patent
Legal protection for a new invention.

Software licence
A legal agreement that sets out how software can be used.

Source code
The program code that software is written in.

Open source software
Software whose source code is available for anyone to see, change and share.

Proprietary software
Software whose source code is kept secret and which users cannot change, usually paid for.

Exam mistakes that lose marks

- **Getting law names or years wrong.** Learn them exactly: The Data Protection Act 2018, Computer Misuse Act 1990, Copyright Designs and Patents Act 1988.
- **Putting an action under the wrong law.** Hacking and malware: Computer Misuse Act 1990. Misusing personal data: The Data Protection Act 2018. Copying software or music: Copyright Designs and Patents Act 1988.
- **Saying the Data Protection Act protects "all data".** It covers personal data about living people.
- **Saying open source just means "free".** Its key feature is that the source code is available and can be changed.
- **Saying proprietary software is always better quality.** Give benefits and drawbacks linked to the scenario.

- **Writing one-sided answers in a "discuss" question.** Give points for and against, cover several of the issues (ethical, legal, cultural, environmental, privacy), and finish with a conclusion.
- **Making points without examples.** Tie every point to the technology or scenario in the question.
- **Mixing up ethical and legal.** Something can be legal but still unethical, such as tracking users in ways they did not expect.

Quick check

1. Give one environmental impact of digital technology.
2. Name the law that makes it illegal to hack into a computer system.
3. State two things The Data Protection Act 2018 requires of organisations that hold personal data.
4. Which law would someone break by copying a film and sharing it online?
5. Give one feature of open source software and one feature of proprietary software.

▼ **Answers**

1. Any one: e-waste from thrown-away devices; energy used by data centres; mining for rare metals; or a positive one such as less travel because of video calls.
2. Computer Misuse Act 1990.
3. Any two: use data fairly, lawfully and openly; use it only for the purpose it was collected for; keep only what is needed; keep it accurate and up to date; keep it no longer than necessary; keep it secure.
4. Copyright Designs and Patents Act 1988.
5. Open source: the source code is available and can be changed. Proprietary: no access to the source code, and it is usually bought off-the-shelf.